

Step-Up Fabrication Document

Top Fabrication (Scale 1:1)

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- 1) FABRICATE PER IPC-6012A CLASS 2.
- 2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

DIMENSIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

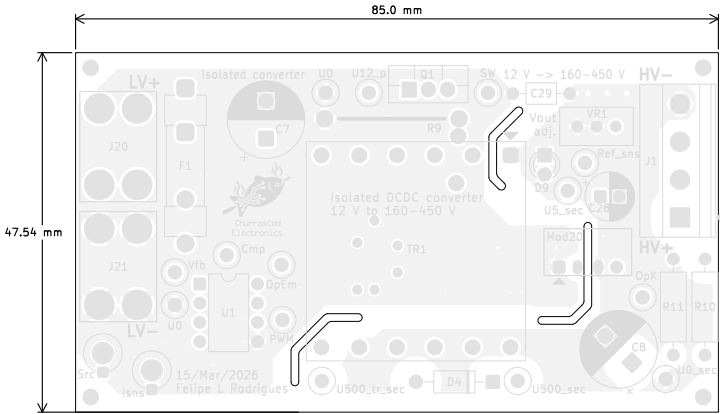
SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 4) SURFACE FINISH: HAL LEAD-FREE
- 5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR YELLOW.
- 6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
- 7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- 8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- 9) PCB MATERIAL REQUIREMENTS:

A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.

B. Tg 170 C OR EQUIVALENT.

C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY BUYER.
- 10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

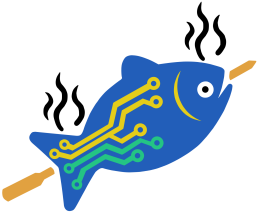
BOARD SIZE	85,000 × 47,540 mm
BOARD THICKNESS	1,600 mm
TRACE WIDTH	0,300 mm
TRACE TO TRACE	0,020 mm
MIN. HOLE (PTH)	0,250 mm
MIN. HOLE (NPTH)	2,100 mm
ANNULAR RING	0,150 mm
COPPER TO HOLE	0,200 mm
COPPER TO EDGE	0,250 mm
HOLE TO HOLE	0,254 mm



Layer Stack Legend

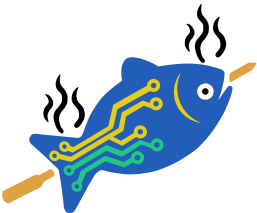
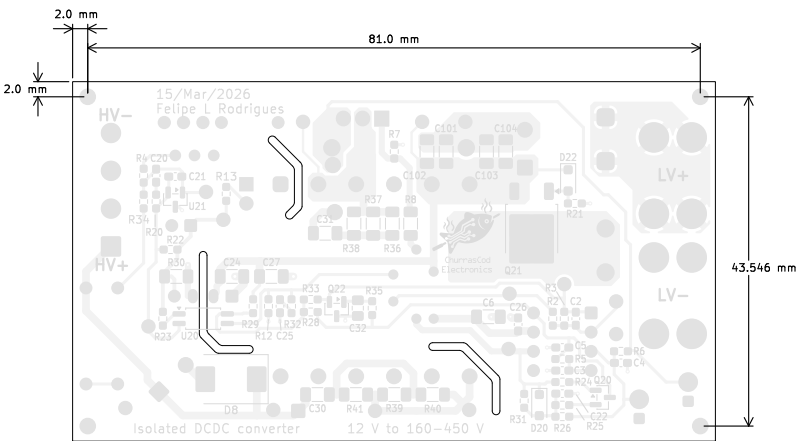
Material	Layer	Thickness	Dielectric	Type	Gerber
	F,Paste			Paste Mask	
	F,Silkscreen		Direct Printing	Legend	GBR
	F,Mask	0,02 mm	Solder Resist	Solder Mask	GBR
Copper	L1 (Sig, PWR)	0,035 mm (1 oz)		Signal	GBR
Core		1,49 mm	FR4	Dielectric	
Copper	L2 (Sig, PWR)	0,035 mm (1 oz)		Signal	GBR
	B,Mask	0,02 mm	Solder Resist	Solder Mask	GBR
	B,Silkscreen		Direct Printing	Legend	GBR
	B,Paste			Paste Mask	

Total thickness: 1,6 mm
Note: external layer thicknesses are specified after plating



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Comments:	Company: NA		Variant: RELEASED	Git Hash: 0689f74
	Board Name: Step-Up		Project Name: Isolated step-up 12:450 V	
Sheet Title: Top Fabrication (Scale 1:1)	File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-07	Revision: 3.0.1
Sheet Path:		Reviewer: FR	Size: A4	Sheet: 1 of 10



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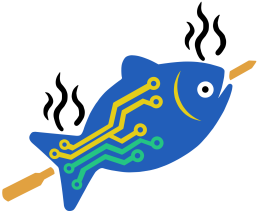
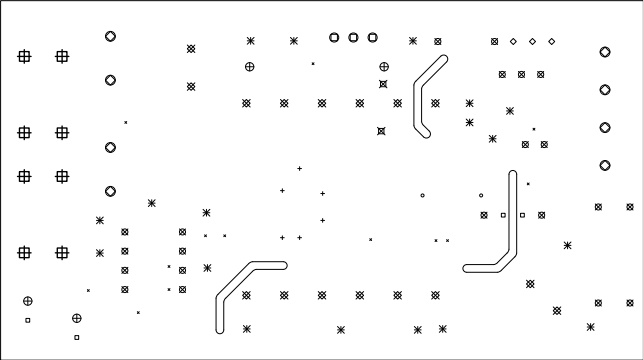
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Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-07	Revision: 3.0.1		
Sheet Path:		Reviewer: FR	Size: A4	Sheet: 2 of 10		

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Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	13	0,25mm (9,84mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Via
○	2	0,40mm (15,75mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Via
+	6	0,50mm (19,69mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Via
□	4	0,70mm (27,56mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
◇	3	0,80mm (31,50mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Via
⊗	21	0,80mm (31,50mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
※	18	0,90mm (35,43mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
⊗	2	1,00mm (39,37mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Via
⊗	16	1,00mm (39,37mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
⊕	4	1,10mm (43,31mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
○	3	1,20mm (47,24mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
○	8	1,30mm (51,18mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
⊕	8	1,80mm (70,87mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
Total 108						



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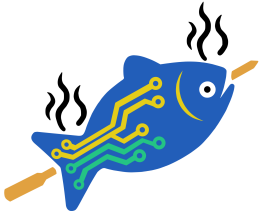
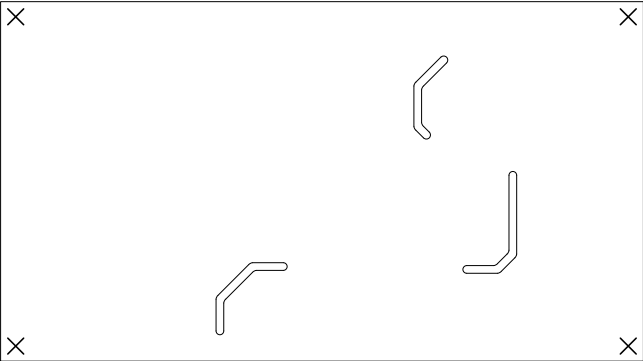
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Sheet Title: Drill Drawing (L1 - L4)	File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-07	Revision: 3.0.1	
Sheet Path:		Reviewer: FR	Size: A4	Sheet: 3 of 10	

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Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	4	2.10mm (82.68mils)	NPTH	Round	L1 (Ref) - L2 (Sig, PWR)	Mechanical
	Total 4					

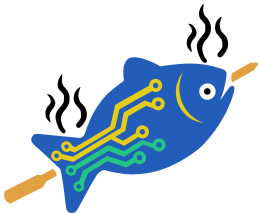
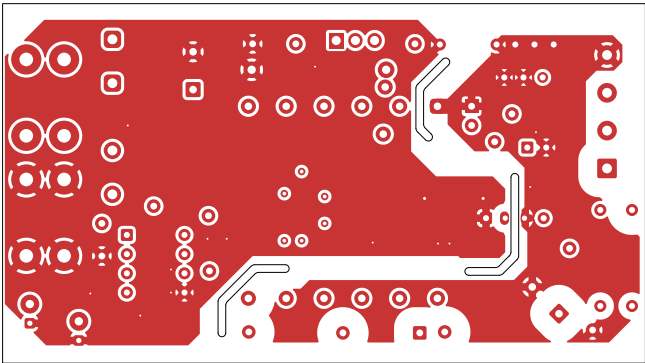


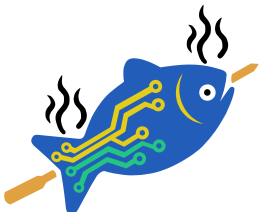
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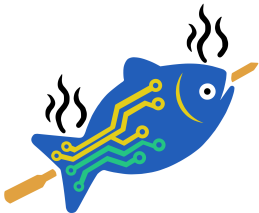
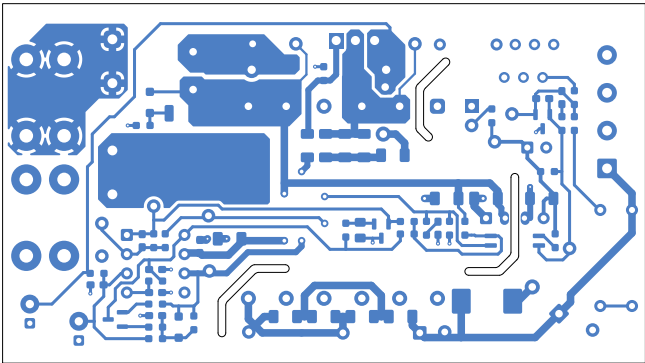
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Sheet Title: Drill Drawing (L1 - L4)	File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-07	Revision: 3.0.1
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F.Cu (Scale 1:1)

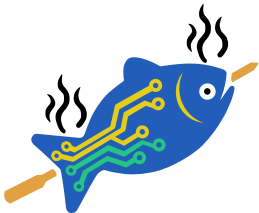
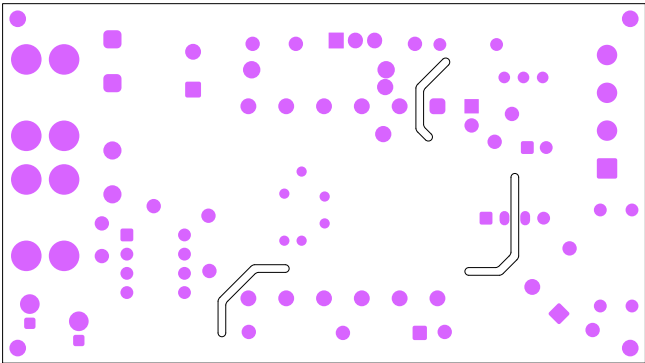


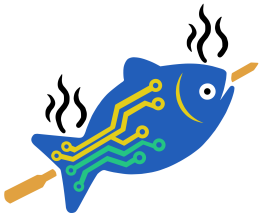
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F.Cu (Scale 1:1)		Isolated_12to450V.kicad_pcb		FR	2026-02-07	3.0.1	
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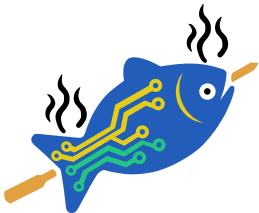
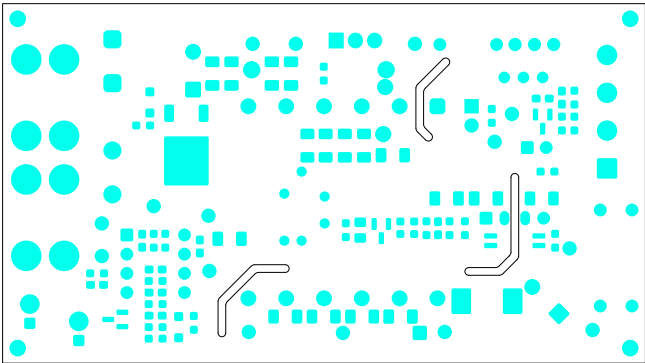


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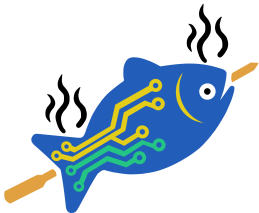
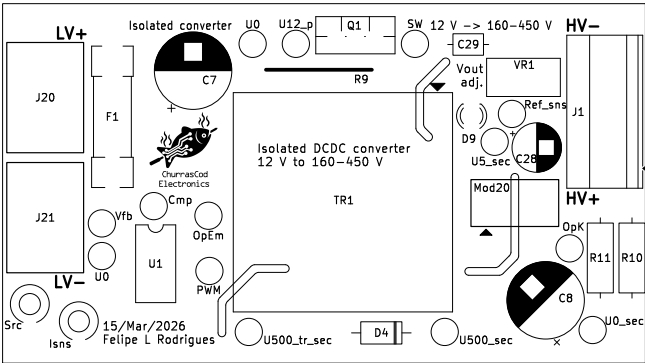


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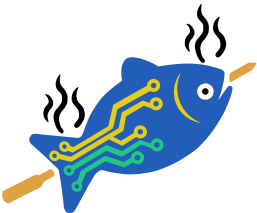
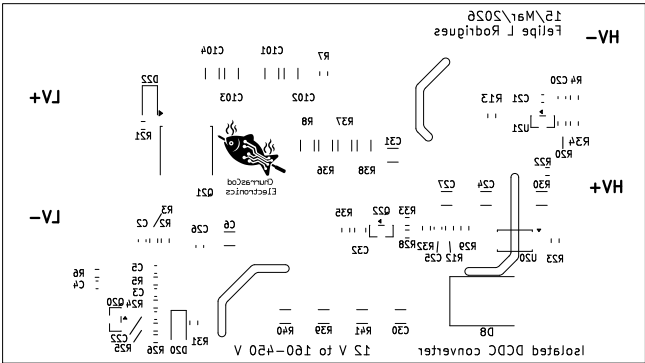
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Sheet Title: B.Mask (Scale 1:1)	File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-07	Revision: 3.0.1
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	Reviewer: FR			Size: A4	Sheet: 9 of 10



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Comments:	Company: NA		Variant: RELEASED	Git Hash: 0689f74
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Sheet Title: B.Silkscreen (Scale 1:1)	File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-07	Revision: 3.0.1
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